
SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting Large Language Models (LLMs) is essential for provenance verification and model attribution. Existing fingerprinting methods are primarily evaluated after fine-tuning, where models have already acquired stable signatures from training data, optimization dynamics, or hyperparameters. However, most of a model’s capacity and knowledge are acquired during pretraining rather than downstream fine-tuning, making large-scale pretraining a more fundamental—and, as our measurements show, a far more demanding—regime for lineage verification. Here, we find that every existing baseline we test becomes *unreliable*: on OLMo-2-7B, all five prior methods fall below the standard 0.8 detection threshold within the first trillion training tokens, because they rely on post-hoc signatures that only emerge after substantial training. This limitation contradicts the classical Galton notion of a fingerprint as an intrinsic and persistent identity. In contrast, we propose a stronger and more intrinsic notion of LLM fingerprinting: **SeedPrints**, a method that leverages random initialization biases as persistent, seed-dependent identifiers present even before training begins. We show that untrained models exhibit reproducible prediction biases induced by their initialization seed, and that these weak signals remain statistically detectable throughout training, enabling high-confidence lineage verification. Where the strongest prior methods drop below the detection threshold during early pretraining and misclassify true descendants under sharp distribution shifts, **SeedPrints** stays verifiable at *every* stage—from initialization, through large-scale pretraining, to downstream adaptation. Across LLaMA-style and Qwen-style models, SeedPrints separates models that differ *only* in initialization seed (cross-seed $p > 0.01$) while confirming true lineage with p -values as small as 10^{-26} , delivering birth-to-lifecycle identity verification that no baseline matches. On large-scale pretraining trajectories and real-world fingerprinting benchmarks, SeedPrints maintains this reliability under prolonged training (up to 700B fine-tuning tokens), domain shifts, and parameter modifications: on the *LeaFBench* suite of 65 models spanning 7 families and 6 transformations, it reaches an overall AUC of 0.992, statistically on par with the strongest baselines (0.994) yet uniquely retaining verifiability throughout pretraining.

Together, these results show that initialization itself imprints a unique and persistent identity on LLMs, forming a true “Galtonian” fingerprint. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints have recently been proposed as a tool to identify, attribute, and trace LLMs by examining their observable behaviors (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). Such methods aim to provide model owners with a verifiable link between a suspicious model and its putative original, enabling detection of model theft or unauthorized reuse (Yoon et al., 2025; Zhang, 2025). The central quantity of interest is a decision: given a base model and a suspect, do they share lineage? A useful fingerprint must therefore render that decision correctly and confidently across the model’s entire lifetime—and, as we show, this is precisely where current methods diverge sharply from ours.

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

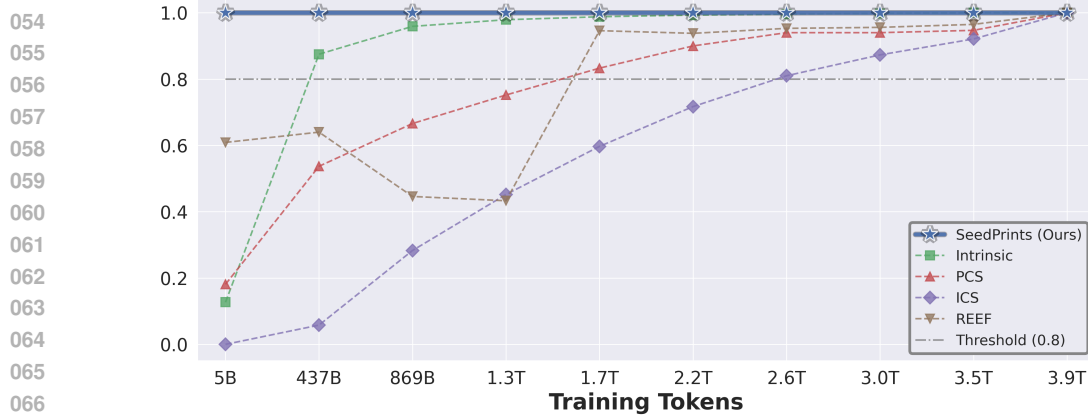


Figure 1: **On OLMo-2-7B, every existing baseline fails to detect lineage during early pre-training, whereas SeedPrints detects it from the first checkpoint.** We compare five methods across ten OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, each tested against the final checkpoint. The y-axis is the similarity score (higher = stronger lineage signal), and the dashed line is the standard 0.8 detection threshold. The contrast is unambiguous: within the first trillion tokens, all four baselines slide below 0.8 (ICS collapses to ≈ 0 , PCS stays under 0.3, and REEF under 0.6), so none of them can verify lineage early in training. **SeedPrints** instead achieves perfect detection ($p \ll 0.001$, plotted as $1 - p$) at every checkpoint from the very first onward, and its signal only strengthens as training proceeds.

Much of this literature explicitly borrows the metaphor of biological fingerprints from Francis Galton’s *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

The analogy suggests that an effective fingerprint should be both unique and persistent, present from the very moment of a model’s “birth” at initialization. Yet most existing so-called fingerprinting approaches fall short of this standard (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025), and the gap is quantitative, not merely conceptual. As shown in Figure 1, all five baselines drop below the 0.8 detection threshold during early pretraining, whereas SeedPrints stays above it throughout. This failure arises because these methods are defined only after models have fully converged, e.g., by extracting patterns from parameters or generated text. As a result, the separability they achieve reflects not a birthmark of the model itself, but an imprint of surface-level training components such as data signatures, optimization dynamics, or hyperparameters. The same fragility surfaces under distribution shift: in Section 5.1, when continued training moves to code data (The Stack), every baseline misclassifies the true descendant, while SeedPrints still returns $p \approx 0$ for the correct lineage. Such methods, therefore, function more as post hoc identifiers than as **Galtonian fingerprints**—those innate, immutable marks that accompany a model from its very beginning.

In this work, we propose a stricter notion of LLM fingerprinting as an intrinsic property present at *initialization* and detectable at *any* time of the subsequent training. We validate this claim with a consistent quantitative pattern across every setting we study—seed-level separability at birth, sub-0.01 p -values throughout training, and benchmark-level parity with the strongest baselines. Our key contributions are:

- **Identifying a fundamental limitation of existing methods.** We show that all five prior fingerprinting approaches fall below the 0.8 detection threshold during early pretraining, and that each is misled under substantial shifts in the training data distribution—misclassifying the true descendant once continued training moves to code (The Stack). (Section 5.1)
- **Discovering and leveraging initialization-driven fingerprints.** We uncover that untrained models exhibit seed-dependent biases in their internal representations, forming a weak but persistent identity signal present from initialization. (Section 3) We then introduce **SeedPrints**, a method that isolates these initialization-born fingerprints and remains more robust to confounding factors such as data distribution, training duration, and optimization dynamics. (Section 4)

- **Extensive empirical validation.** We demonstrate that our method separates models differing *only* in initialization seed—even under identical training pipelines and data orders, where every cross-seed test yields $p > 0.01$ —while confirming true init-to-base lineage with p -values as small as 10^{-26} . These signals persist under continual training across diverse datasets. (Section 5.1)
- **Strong performance in realistic settings.** Our method reliably detects lineage throughout large-scale pretraining trajectories (e.g., OLMo-2-7B Stage 1) with $p \ll 0.01$ at every checkpoint, avoiding the early-stage collapse of all baselines. It also holds $p < 0.01$ across large-scale fine-tuning (5M up to 700B tokens) and practical deployment scenarios on the *LeaFBench* benchmark (Shao et al., 2025), which spans 65 models, 7 mainstream model families, and 6 transformations (instruction tuning, fine-tuning, PEFT, quantization, model merging, and distillation). Here SeedPrints reaches an overall AUC of 0.992 and KS of 0.986—statistically on par with the strongest baselines (0.994/0.989) and far ahead of weaker ones such as Gradient (0.801)—while remaining the only method verifiable throughout pretraining (Sections 5.2 and 5.3).

2 RELATED WORK

LLM protection broadly falls into two families: (i) *watermarking / active fingerprinting* methods that *insert* an identifiable signature into a model or its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* methods that *extract* a signature from a model’s pre-existing behaviors without modifying it (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025).

Watermarking and Active Fingerprinting. Active approaches deliberately implant verifiable identifiers for later ownership checks. Classic techniques include backdoor attacks (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures and hash functions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two common forms are: *text watermarks*, which bias generation or insert predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.); and *model weight watermarks*, which embed identifiers into parameters or link them to secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are relatively straightforward and may persist after moderate fine-tuning (e.g., Dasgupta et al. 2024), these invasive schemes require control of the training process, making them unsuitable for retroactively marking third-party models.

Passive LLM Fingerprinting. In contrast, passive fingerprinting identifies models by analyzing their intrinsic properties without any modification. Passive fingerprinting techniques vary by model access. With **white-box access**, signatures are extracted from model weights, leveraging intrinsic properties like the distribution of attention matrices (Yoon et al., 2025), the kernel alignment of internal representations (Zhang et al., 2024), or the stable direction of parameter vectors. In the **black-box setting**, fingerprinting relies on analyzing input-output behavior. These methods use crafted queries (Pasquini et al., 2024), unique prompt-response pairs (Tsai et al., 2025), stylometry (Alhazbi et al., 2025) or iterative prompting–response games (Iourovitski et al., 2024) to identify a model, though they can be less robust to fine-tuning. However, these methods define their signatures *post hoc*. Specifically, they identify emergent properties from a completed training process, rather than the innate, “*Galtonian*” fingerprints present from random initialization that our work seeks to discover. As a result, *they are not designed for lineage verification during large-scale pretraining*, where such post-training signals have not yet emerged—an evaluation gap our experiments make concrete, with all five representative white- and black-box baselines falling below the 0.8 detection threshold within the first trillion pretraining tokens.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

In this section, we present our key observation that language models exhibit strong prediction biases originating from the random initialization seed, and—crucially for a fingerprint—that these biases remain measurably detectable even after training. Two quantities anchor this observation: the *shape* of the initialization bias (how far outputs depart from uniform) and its *persistence* (how much of it survives training), and we characterize both below.

Initialization induces seed-specific bias profiles. We evaluate a LLaMA-2–style model initialized with seed 123 on 10,000 random input sequences of length 1,024, where tokens are sampled uniformly from the vocabulary. In Figure 2 (Left), we plot the frequency with which each output token (upper, from the final logits) and each hidden dimension (lower, from the last-layer representations) attains

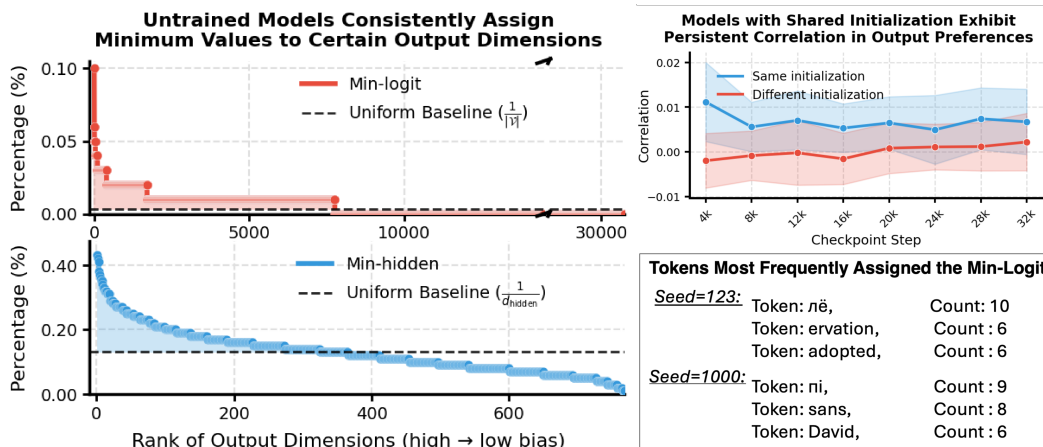


Figure 2: **Initialization-born output bias persists through training—and separates same-seed from independent models.** *Left:* Under completely random inputs, a randomly initialized LLaMA-2-style model produces outputs far from uniform: a few coordinates absorb most of the minimum values (the dashed line marks the uniform-distribution expectation these frequencies far exceed). The same extreme bias appears in both the logits (top, red) and the final hidden representations (bottom, blue). Arrows in the top panel denote a broken x-axis omitting low-frequency tail ranks. *Upper Right:* Throughout training, a model stays weakly but consistently correlated with the base model it was trained from. The key comparison is between the two distributions: same-initialization correlations sit systematically *above* those computed against independently initialized base models, so the initialization signal remains measurable rather than washing out. *Lower Right:* Which dimensions carry the strongest bias is seed-specific—illustrated here with the tokens most frequently receiving the minimum logit—confirming that the persistent signal is an identifier, not a generic artifact.

the minimum value across the 10,000 random trials. Both plots depart sharply from the uniform-frequency baseline: a small set of coordinates absorbs a disproportionate share of the minima, exposing an extreme, structured output-preference bias rather than random noise. Repeating the experiment across different initialization seeds isolates what is generic from what is identifying: the overall *magnitude* of bias is stable across seeds, but the specific argmin dimensions shift with the seed (Figure 2, *Lower Right*)—exactly the property a seed-level fingerprint requires. These observations are consistent with recent findings that biases in randomly initialized models arise from inter-sequence representation contraction, driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026). Under this view, different initialization seeds induce different contraction directions, resulting in distinct argmin patterns across output dimensions.

Training preserves initialization-born bias. Although training substantially changes output magnitudes, the relative preference over output dimensions induced at initialization is not erased—and the residual signal is directional, not random. We train the previously initialized model (seed 123) for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate intermediate checkpoints. We focus on the $m = 50$ output dimensions most frequently assigned minimum values by the initialized model. For each checkpoint, we compute the correlation between its responses and those of the initialized model across random inputs, restricted to these dimensions; intuitively, this measures how similarly the checkpoint and its initialization rank or disfavor these dimensions across inputs. The decisive comparison is against a matched control: the same correlation computed against an independently initialized model (seed 1000). As shown in Figure 2 (Upper Right), the two distributions separate. Although the absolute correlations are small, same-initialization correlations are consistently shifted *above* the independent-baseline distribution and, tellingly, *stabilize* rather than decaying toward zero as training proceeds. This gap—small in magnitude but consistent in sign and persistent over training—is exactly the weak-but-statistically-detectable signal our test in Section 4 is built to amplify.

4 ALGORITHM

Section 3 established the empirical foundation we now exploit: models from the same initialization lineage retain a weak but consistent agreement on a subset of output dimensions (those most disfavored by the untrained model under random inputs), and this agreement is measurably larger than for unrelated models. Our goal is therefore to recover, from any two given models (without access to the

true initialized model), the dimensions that carry this bias signal and to test whether their agreement is statistically significant—turning the small distributional gap of Figure 2 into a calibrated p -value. The overall procedure is summarized in Algorithm 1, and we describe each component below.

Extract identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . Each random input x_i can be instantiated either by uniformly sampling ℓ token IDs from the vocabulary, or by directly sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality. Here, $g(x_i)$ can denote either the model’s logits or its final-layer representations. We then identify its disfavored dimensions as the set of bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We use the mean response vector to extract bias signals, as it provides a more stable and noise-robust alternative to per-sample argmin frequency (i.e., counting how often each dimension attains the minimum value, as illustrated in Figure 2), while also performing well empirically. We provide a formal comparison in Appendix C.1.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. Intuitively, if two models share the same initialization lineage, they should independently identify a similar set of disfavored dimensions, leading to non-trivial overlap in their bottom- m sets. This intersection-based criterion therefore acts as a form of mutual verification: it suppresses spurious correlations arising from architectural similarity or shared training dynamics, and isolates bias signals that are more likely to originate from shared initialization lineage. When one of the models is closer to initialization, its disfavored dimensions can also be used directly as an anchor.

Correlation statistics. Restricting both models to the identity dimensions $\mathcal{S}$, we compare their responses across random inputs.¹ For each $j \in \mathcal{S}$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This yields a set of correlation statistics $T = \{\tau_j\}_{j \in \mathcal{S}}$, which captures the distribution of agreement between the two models across identity dimensions. As observed in Section 3, models from the same lineage exhibit distributionally higher agreement on these dimensions compared to unrelated models.

Hypothesis testing via null distribution. Since the correlation distributions for same-lineage and different-lineage models can overlap, a simple threshold on individual statistics is insufficient. Instead, we test whether the observed correlations are significantly larger than what would be expected under the null hypothesis of no lineage relation. Ideally, this null should be estimated from the correlation distribution between independent models as in Figure 2. However, obtaining such a baseline empirically requires comparisons across many independently initialized models, which is computationally expensive. As shown in Appendix B, the null correlation distribution between independently initialized models is well approximated by a Gaussian distribution. This naturally motivates a cheaper simulation-based alternative: constructing an empirical null distribution using Gaussian surrogates passed through the same pipeline, as summarized in Algorithm 1.

While this simulation-based null already provides a strong empirical approximation, it still introduces unnecessary randomness and computational overhead: its precision is limited by the number of simulation trials, results can vary across runs, and comparing two empirical distributions typically requires additional assumptions from the adopted test, such as the one-sided t-test or the Mann–Whitney U test. Since the null distribution of Kendall–Tau (under weak independence) is available in closed form, we further replace this simulation-based procedure with an analytical test, while also providing both empirical and analytical implementations in our code repository. Specifically, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|\mathcal{S}|$ increases, the distribution of $\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j$ is well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right)$. While softmax

¹Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{S}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

normalization introduces weak dependencies across dimensions, these effects are mitigated by the use of a relatively high temperature ($T = 10$), making the independence assumption a reasonable approximation in practice. We empirically validate that this Gaussian approximation closely matches the distribution obtained from the full pipeline (Appendix B). We therefore compute the one-sided z-score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|\mathcal{S}|}}$, and the corresponding p-value $p = 1 - \Phi(z)$, where Φ is the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage if $p < 0.01$.

Practical instantiation. Model outputs can be taken as either logits or hidden representations, and both are consistent with our formulation. The choice, however, has a concrete cost: because the logit space grows with the vocabulary, reliably estimating its bias pattern demands substantially more random inputs, inflating compute (see Appendix C.2 for the sample-complexity analysis). We therefore default to final-layer hidden states, which deliver the same test more efficiently and stably. The decisive practical advantage is what the test outputs: a calibrated p -value that supports a direct accept/reject decision on shared lineage, rather than a heuristic similarity score that must be compared against a hand-tuned threshold.

5 EXPERIMENTS

This section is organized around two claims, each backed by a direct comparison against the same set of baselines. In Section 5.1, we show that our method functions as a genuine fingerprint on two axes where prior methods fail: (i) *birth verification at the seed level*—separating models that differ only in seed (all cross-seed $p > 0.01$) while confirming true init-to-base lineage at $p < 0.01$ —and (ii) *verifiability across the entire training lifecycle*. By contrast, the baselines cannot verify lineage at early pretraining and, relying on data-dependent signals, break down under significant distribution shifts. Unless noted, we report the analytical null with *hidden state* outputs; the Appendix D.2 adds empirical-null results for both *logit* and *hidden state* outputs under the one-sided t -test and Mann–Whitney U test, with consistent conclusions. In Section 5.3, we stress-test all methods under practical infringement scenarios using *LeaFBench* (Shao et al., 2025). Across 65 distinct model instances spanning 6 representative post-development techniques, SeedPrints matches the best baselines in post-training settings (overall AUC 0.992 vs. 0.994) while remaining robust to every deployment transformation.

Baselines. We compare against four passive fingerprinting baselines (weight- or representation-based), spanning the two dominant white-box families. *Intrinsic fingerprint* (Yoon et al., 2025) (or *PDF* in some papers) compares models via the similarity of the layerwise standard-deviation profiles of attention parameters. *REEF* (Zhang et al., 2024) computes centered-kernel-alignment (CKA) similarity between feature representations from the same samples across two models. *PCS* and *ICS* (Zeng et al., 2024) (collectively *HuRef* in some papers) are weight-similarity methods: PCS flattens all parameters and measures cosine similarity, while ICS forms invariant terms from the weights and measures cosine similarity on those invariants. In Section 5.3, we additionally include a (weaker) gradient-based fingerprint *Gradient* (Shao et al., 2025). For fair comparison, all four baselines emit similarity scores and, following Zhang et al. (2024), are thresholded at **0.8** for binary decisions; SeedPrints instead emits a calibrated p -value thresholded at $\alpha=0.01$. Keeping this

Table 1: Seed-level separability. Table 2: Fingerprint persistence through training. SeedPrints Every cross-seed pair yields $p > 0.01$ (all correct), so distinct initialization seeds are never falsely merged.

Model Pairs	p -value (> 0.01)	Model Pairs	SeedPrints p -value	Baselines			
				Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{2000}^{base}	0.116	s_{42}^{init} vs. s_{42}^{base}	3.37e-26✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{42}^{base}	1.000	s_{123}^{init} vs. s_{123}^{base}	3.16e-33✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{123}^{base}	0.327	s_{1000}^{init} vs. s_{1000}^{base}	1.12e-21✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{1000}^{base}	1.000	s_{2000}^{init} vs. s_{2000}^{base}	3.70e-31✓	-0.337×	0.331×	0.581×	0.188×

Table 3: Seed is the sole driver. With corpus and data order held identical, all cross-seed pairs still read $p > 0.01$ (0.109–0.883), far from the 10^{-21} – 10^{-33} of true lineage.

Model Pair	p -value (> 0.01)	Setting	SeedPrints p -value	Baselines			
				Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{123}^{base}	0.573	Continual corpus (seed)					
s_{123}^{init} vs. s_{2000}^{base}	0.724	TinyStories (1000)	≈ 0 ✓	1.000✓	0.759×	0.999✓	0.996✓
s_{2000}^{init} vs. s_{1000}^{base}	0.109	TinyStories (123)	1.000✓	0.950×	0.658✓	0.332✓	0.012✓
s_{1000}^{init} vs. s_{42}^{base}	0.883	the_stack (1000)	≈ 0 ✓	0.489×	0.557×	0.585×	0.123×
		the_stack (123)	1.000✓	0.445✓	0.580✓	0.301✓	0.026✓

contrast—similarity-plus-threshold vs. a direct statistical test—in view is essential to reading the tables that follow.

To make the comparison structure legible at a glance, all experiment tables share one color convention: green marks a same-source (true-lineage) pair whose correct verdict is “share lineage,” and red marks a different-source pair whose correct verdict is “distinct.” For example, s_{42}^{init} vs. s_{42}^{base} pairs a seed-42 initialization with its own continued-pretrained descendant (green), whereas s_{2000}^{init} vs. s_{42}^{base} pairs a seed-2000 initialization against a model trained from a seed-42 initialization (red). A ✓ therefore marks a correct decision and a × an error, so a method’s reliability can be read off directly from how many checkmarks its column carries.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper reports LLaMA-style results and defers the Qwen-style counterparts to Appendix D; the two families yield the same quantitative pattern throughout.

Different initialization seeds produce distinct fingerprints. Table 1 reports p -values from our correlation test on every pair of models initialized with different random seeds (42, 123, 1000, 2000). All four cross-seed p -values stay above the 0.01 threshold (ranging from 0.116 up to 1.000), so not a single distinct-seed pair is falsely merged. Distinct seeds thus produce distinct fingerprint behaviors, letting models be separated “at birth.”

Training preserves the initialization fingerprint. Table 2 pairs each initialization model s_i^{init} with its own descendant s_i^{base} , continued-pretrained on OpenWebText (Gokaslan et al., 2019) (≈ 10 B tokens). The contrast with the cross-seed case is stark: SeedPrints returns p -values between $3.16\text{e}-33$ and $1.12\text{e}-21$ for all four true-lineage pairs—dozens of orders of magnitude below 0.01—so every pair is correctly attributed ($\checkmark \times 4$). The trained model plainly inherits its initialization’s fingerprint. The same table exposes the baselines’ blind spot: across all four pairs, Intrinsic, REEF, PCS, and ICS each fall on the wrong side of the 0.8 threshold (16/16 errors), confirming that whatever separability they achieve is an artifact of training rather than a signal of initialization lineage.

Identical data and order do not make fingerprints converge. Table 3 isolates the initialization seed as the sole source of variation: all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$

are trained on *exactly the same corpus* (OpenWebText) and in the same data order (the training seed is fixed to lock the curriculum), differing only in their initialization seed i . Even under this maximally confounding setup, every cross-seed p -value stays above the threshold (0.109–0.883)—orders of magnitude apart from the 10^{-21} to 10^{-33} values recorded for true lineage in Table 2. The fingerprint is therefore driven by the seed itself, not by shared data or training order.

Continual training on diverse datasets does not confound the fingerprint. The seed-level experiments above establish that SeedPrints behaves as a biometric fingerprint; from a copyright standpoint, a baseline’s inability to tell seeds apart is not itself a fault, since seeds carry no clear copyright status. The consequential failure is different and measurable: baseline attribution is *misled by data distribution, losing the true lineage when the training distribution shifts substantially during continued training*.

Table 4 makes this precise. We continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two deliberately dissimilar corpora—TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code)—and ask each method to pick the true descendant over a *distractor* derived from a different base (seed 123, pretrained with a different data order on OpenWebText and then continued on the same target corpus).

The results split cleanly by domain. On the near-domain corpus TinyStories, most methods succeed; but on the far-domain corpus The Stack, *all four baselines misclassify the true descendant* (Intrinsic 0.489, REEF 0.557, PCS 0.585, ICS 0.123—every score below the 0.8 threshold for a pair that should read as same-lineage). This is the signature of methods tracking *domain similarity* rather than lineage: TinyStories stays close to the OpenWebText pretraining distribution, whereas The Stack diverges sharply enough to slip past detection. SeedPrints, by contrast, returns $p \approx 0$ for both true descendants and $p = 1.000$ for both distractors—correct on all four cases regardless of corpus. The fingerprint is thus no proxy for data distribution: it survives a large domain shift and persists well beyond the initial pretraining stage.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. Can our method track a model’s offspring along the entire training trajectory? We first answer this in a controlled OpenWebText run with known initialization seed, testing lineage at every intermediate checkpoint. Detection is uniform: all checkpoints are verified with $p < 0.01$, without a single early-stage miss (Appendix D.2.1, Figure 5).

Large-scale pretraining. We then move to realistic foundation-model training, analyzing ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens), treating the final checkpoint as the target and all earlier checkpoints as tested ancestors. Unlike the controlled setting, this trajectory spans heterogeneous corpora, far longer optimization, and large-scale representation drift—the conditions under which prior methods are expected to be weakest. For directional comparability with similarity-based baselines we plot $1 - p$; across all ten checkpoints SeedPrints yields $p \ll 0.01$, numerically indistinguishable from zero.

The comparison in Figure 1 is decisive. **Every baseline fails to verify lineage during early pretraining**—within the first trillion tokens ICS collapses to ≈ 0 , PCS stays below 0.3, and REEF below 0.6, all under the 0.8 threshold—whereas SeedPrints is verifiable from the very first checkpoint and only strengthens thereafter. This exposes a critical gap in current evaluation practice: **prior work overwhelmingly verifies lineage only after fine-tuning, where detection is substantially easier**. Because large-scale pretraining itself reshapes representations and inflates late-stage lineage signals, evaluating only at convergence can create a *false sense of safety*. We argue that verification must be judged where it is hardest—the early pretraining stages, precisely when misuse is most difficult to detect—rather than on late checkpoints alone.

Large-scale finetuning. We next evaluate the standard, easier regime—lineage after fine-tuning—to show SeedPrints loses nothing where baselines are strong. Suspects are six diverse downstream variants of Llama-2-7B spanning fine-tuning volumes from 5M to 700B tokens: Llama-2-finance-7b (Heenan, 2023) (5M), Vicuna-1.5-7b (Chiang et al., 2023) (370M), WizardMath-7b (Luo et al., 2023) (1.8B), Chinese-LLaMA-2-7b (Chen et al., 2023) (13B), Code-Llama-7b (Meta AI, 2024) (500B), and Llemma-7b (Azerbayev et al., 2023) (700B). As shown in Table 5, SeedPrints holds $p < 0.01$ on all six—by margins so large (10^{-3552} even at 500B tokens) that the decision is never

Table 5: Large-scale fine-tuning, each row a target vs. LLaMA-2-7B, ordered by fine-tuning volume (5M→700B tokens). **SeedPrints** reports the correlation-test p -value (< 0.01 = strong signal); the four baselines report similarity scores (threshold 0.8, higher = better). SeedPrints verifies all six targets with overwhelming margins ($p \leq 10^{-3552}$ throughout), whereas PCS and ICS drop below 0.8 and misclassify the two largest-budget models (Code-Llama, Llemma).

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000✓	0.9950✓	0.9979✓	0.9952✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000✓	0.9985✓	0.9985✓	0.9949✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000✓	0.9979✓	1.0000✓	0.9994✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990✓	0.9978✓	1.0000✓	0.9817✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480✓	0.9947✓	0.6863✗	0.3369✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470✗	0.9984✓	0.6682✗	0.2905✗

in doubt. Notably, at the two largest fine-tuning budgets the weight-space baselines PCS and ICS both drop below 0.8 (PCS 0.686/0.668, ICS 0.337/0.291) and misclassify Code-Llama and Llemma, whereas SeedPrints remains decisive.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely apply *parameter-altering* adaptations to foundation models (e.g., fine-tuning, PEFT, quantization) that can weaken or distort fingerprint signals. To quantify reliability under these conditions, we adopt *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing, and report head-to-head AUC/KS against all baselines. The headline result: SeedPrints reaches an overall AUC of 0.992, statistically tied with the two strongest baselines (Intrinsic and ICS at 0.994) and well clear of REEF (0.915) and Gradient (0.801)—while being the only method that also survives early pretraining.

Across Source Models. We consider two source-model types for auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each, the task is a binary lineage decision—separate *derivative* models (obtained by post-training adaptation of the source) from *independent* models (trained without the source weights)—mirroring the two auditing scenarios operators actually face. The evaluation covers 65 models in total; the full list of names and HuggingFace repositories is in Appendix E.

Metrics. We report two complementary detection metrics: (i) the Area Under the ROC Curve (AUC), summarizing separability across all thresholds, and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), the maximum gap between the score distributions of derivative and independent models—a threshold-free measure of how far apart the two populations sit.

One caveat matters for reading these numbers in our favor. SeedPrints emits per-model p -values, which are fundamentally different objects from the baselines’ similarity scores: similarity is a linear scale on which differences are proportional, whereas a p -value is a tail probability that cannot be read linearly. To fit the baselines’ AUC pipeline, we map p -values to scores via $s = 1 - p$. This mapping is deliberately conservative—it compresses p -values that vary on an exponential scale onto a linear threshold sweep, blurring exactly the fine-grained separations SeedPrints is best at—so any AUC we report understates our method’s true discriminative power. Even under this handicap, SeedPrints remains comparable to or better than the baselines. Concretely, we (i) compute AUC for all methods by sweeping thresholds over s , and (ii) report KS as a threshold-free separability measure. Manhattan distance is omitted, as absolute distances between p -values carry no meaning.

This distinction is decisive in practice, where calibrating an optimal similarity threshold is usually infeasible. AUC rewards a method for being separable *after* an oracle picks the best cutoff; SeedPrints does not need that oracle. **For any model pair, it outputs a p -value that yields a definitive same-lineage decision at a fixed significance level, with no tunable threshold**—so its strong AUC understates a further, threshold-free advantage that the baselines cannot claim.

Parameter-Altering Techniques. To assess fingerprinting robustness, we evaluate models produced via (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

Table 6: LeaFBench detection across source models (AUC/KS, higher = better). “PT”/“IT” denote pre-trained and instruction-tuned sources; per-family columns cover the 6 families with both PT and IT models. The “Overall” column pools all test pairs under one global threshold and is thus not the average of the per-family columns. On this global metric SeedPrints (0.992 AUC / 0.986 KS) is statistically level with the best baselines, Intrinsic and ICS (both 0.994 AUC; 0.989/0.943 KS), and far ahead of REEF (0.915/0.739) and Gradient (0.801/0.508).

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC ↑	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic ↑	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC ↑	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic ↑	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC ↑	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic ↑	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC ↑	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic ↑	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC ↑	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic ↑	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

5.3.1 FAMILY-WISE EFFECTIVENESS

Of the seven families, we score six that provide both a PT and an IT source (each family’s PT and IT models are treated as separate sources and compared against the full pool of 65); TinyLlama-1.1B is omitted from Table 6 as a trivial case for every method. Each per-family column is an AUC over that family’s tests with a family-calibrated threshold, whereas the “Overall” column pools all test pairs under a single global threshold—so the overall score is *not* the average of the per-family columns and is the more demanding number. On that global metric, three methods are effectively saturated and statistically indistinguishable: *Intrinsic* and *ICS* at AUC 0.994 and *SeedPrints* at 0.992 (KS 0.989, 0.943, and 0.986, respectively), with most per-family entries hitting the ceiling of 1.000. The remaining baselines trail clearly—REEF at 0.915 AUC / 0.739 KS and Gradient at 0.801 / 0.508. In short, a method built to capture *intrinsic seed-level* fingerprints nonetheless matches the best post-hoc detectors for identity tracking across diverse families, while carrying the added guarantee of early-pretraining verifiability that they lack.

5.3.2 ALTERATION-WISE ROBUSTNESS

Breaking performance down by transformation clarifies where difficulty actually lies. *Instruct* is the easiest case for everyone—even the weakest baseline, Gradient, reaches AUC 0.895—while *Finetune* and *Merge* are the hardest, with no method achieving a perfect score on *Finetune*. The remaining transformations separate the methods sharply: REEF drops from 0.996 on *Instruct* to 0.871 under *Quantization* and 0.858 under *Distillation* (though it stays resilient to *PEFT* at 0.994), and Gradient falls to 0.776/0.756 under *PEFT/Quantization*. Against this backdrop SeedPrints is uniformly strong: it scores 1.000 AUC on *Instruct*, *Quantization*, and *Distillation*, 0.995 on *Finetune* (the single best result on the hardest “easy” case), and 0.990 on *PEFT*. Its only dip is *Merge* (0.959), traceable to exactly one misclassified model out of seventeen—consistent with weight-space interpolation against unrelated weights occasionally diluting the initialization signal below threshold. Averaged over all six transformations the three top methods finish within 0.006 AUC of one another (Intrinsic 0.997, ICS 0.995, SeedPrints 0.991), confirming that SeedPrints delivers essentially best-in-class deployment robustness.

5.4 ABLATION STUDY ON HYPERPARAMETERS

Our method exposes three hyperparameters: the number of random input sequences n , the sequence length l , and the fingerprint size m (how many low-mean coordinates enter the test); n and l jointly set the amount of random input. The main experiments in Sections 5.2 and 5.3 use $n=2000$, $l=1024$, and $m=0.1d_{\text{out}}$ (d_{out} = hidden-state dimensionality), a setting that scales across open-source models. To measure sensitivity, we vary one hyperparameter at a time about the defaults ($n=2000$, $m=400$, $l=768$) and report accuracy and false-positive rate at a fixed significance level $\alpha=0.01$ in Table 8. Two trends stand out: more input monotonically helps—accuracy climbs from 0.7368 to 0.9375

Table 7: LeaFBench detection by parameter-altering transformation (AUC/KS, higher = better). SeedPrints scores 1.000 AUC on Instruct, Quantization, and Distillation and leads on the hard Finetune case (0.995); its only dip is Merge (0.959, one of seventeen models). Averaged over all six transformations the three top methods finish within 0.006 AUC (Intrinsic 0.997, ICS 0.995, SeedPrints 0.991), while REEF (0.924) and Gradient (0.812) trail.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

as n grows $200 \rightarrow 2000$, and from 0.9048 to a perfect 1.0000 as l grows $512 \rightarrow 1024$ —and the false-positive rate stays at or near 0 throughout, so gains never come at the cost of spurious matches.

Table 8: Hyperparameter ablation at significance level $\alpha = 0.01$, varying one of n, l, m about the defaults. Accuracy (ACC) rises with more input (n, l) while the empirical false-positive rate (FPR) stays at 0 except at the extremes ($l=512, m=800$).

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

The monotone gains from n and l are exactly what our theory in Section C.2 predicts: accurate recovery of the population-level bias needs $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$ queries, where γ is the margin separating biased coordinates from the rest. A too-small n therefore falls short of this bound—explaining the 0.7368 accuracy at $n=200$ —while γ itself grows with l through accumulation, so longer sequences enlarge the margin and, at fixed n , lower the query count required for reliable detection.

The fingerprint size m behaves non-monotonically, and the table pins down the sweet spot. Too small an m leaves the selected coordinates unstable for want of observations; too large an m admits irrelevant coordinates that shrink γ and would demand quadratically more samples to offset—visible as accuracy falling to 0.875 and the false-positive rate jumping to 0.1463 at $m=800$. In between, $m \in [200, 400]$ holds accuracy at 0.9375 with a 0 false-positive rate, matching our $m \approx 0.1d_{\text{out}}$ heuristic for the typical range $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

We introduced *SeedPrints*, a stronger, intrinsic notion of LLM fingerprinting that traces a model’s lineage back to its random initialization, and backed it with a consistent quantitative story. Untrained models already carry stable, seed-dependent token-preference patterns, and these biases persist—statistically and measurably—throughout training; a distribution-correlation test over *identity dimensions* converts that weak signal into a calibrated decision. The evidence is uniform across settings: SeedPrints separates seed-only variants (all cross-seed $p > 0.01$) yet confirms true lineage at p as small as 10^{-26} , verifies every OLMo-2-7B checkpoint from 5B to 3.9T tokens where all baselines fall below the 0.8 threshold within the first trillion, holds $p < 0.01$ under fine-tuning up to 700B tokens and under sharp domain shift to code, and reaches 0.992 overall AUC on the 65-model *LeaFBench* suite—statistically level with the strongest baselines (0.994) while alone remaining

verifiable throughout pretraining. Together these results make SeedPrints a practical, all-stage tool for provenance and copyright auditing.

Limitations and Future Directions This work aims to uncover fingerprint signals rooted in initialization-stage bias. The proposed method requires access to model outputs (e.g., logits or hidden states) and may not apply when only restricted API access is available. However, we stress that the pursuit of *true fingerprints* for LLMs is still at an early stage: most existing approaches (especially black-box methods) rely on signals that emerge in later training stages or from alignment-driven artifacts and therefore cannot robustly provide protection at the pretrained stage. Therefore, reliable all-stage fingerprints are even more urgently needed. See full discussions in Section [A.2](#).

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A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We use AI assistants, i.e., ChatGPT and Gemini, for writing and formatting support. Their use covers grammar and style checks, improving clarity of figure and table captions, and other surface-level edits. For programming-related tasks, we occasionally use GitHub Copilot and Claude as coding assistants, e.g., for code auto-completion and debugging hints.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although this work uncovers seed-born, training-persistent biases that serve as fingerprint signals—verifiable from initialization through 3.9T pretraining tokens and 700B fine-tuning tokens—our method remains within the white-box paradigm, requiring access to logits or hidden states. It may not apply under restricted API access, e.g., when responses are deterministic or safety-filtered. We stress, however, that the pursuit of *true fingerprints* for LLMs is still at an early stage even in the white-box setting: most existing approaches rely on training-induced signatures or alignment behaviors and, as our lineage-verification comparisons quantify, all fall below the 0.8 detection threshold within the first trillion pretraining tokens, offering no robust protection for pretrained or early-stage models. By analogy to human fingerprints, which remain stable regardless of age or environment, a true LLM fingerprint should persist across continued training and deployment rather than reflect incidental artifacts. Closing the remaining gap—particularly under black-box access, where a substantial performance shortfall relative to white-box methods persists—is a challenging open direction. We hope this work motivates further research toward principled, robust, all-stage fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

Our hypothesis test in Section 4 evaluates whether two models share lineage by comparing the distribution of their correlation statistics over random inputs against a null distribution. The null distribution models the correlation statistics between two *independent models evaluated on random inputs*.

We consider two approaches for approximating this null distribution: (i) empirical simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. In this section, we justify these approximations through three steps: (i) characterizing the true null using independently initialized models, (ii) introducing the Gaussian surrogate as a tractable simulation-based proxy, and (iii) validating the analytical approximation.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null corresponds to the correlation statistics between two independent models evaluated on random inputs, and the cleanest way to realize that independence is to compare two independently initialized models (different seeds, no training). We estimate this empirical null from a large sample— ~ 2500 such pairs, each scored with Kendall–Tau correlations over 10,000 random inputs (as in Section 3). Figure 3 shows the payoff of this scale: the resulting null is, to the eye, a Gaussian centered at zero, exactly the shape our analytical test assumes.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Directly estimating the null distribution using independently initialized models is computationally expensive, as it requires access to many models and repeated evaluations. To obtain a tractable alternative, we introduce a simulation-based null by using Gaussian surrogates as proxies for model outputs under the null.

This choice is motivated by two considerations. First, approximating neural network outputs with Gaussian distributions is a widely adopted practice in auditing literature, supported by both empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends only

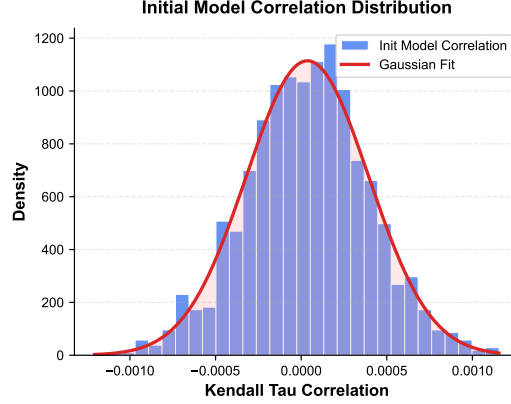


Figure 3: Across ~ 2500 pairs of independently initialized models evaluated on random inputs, the empirical correlation distribution closely tracks a zero-centered Gaussian—validating the analytical null used throughout.

on relative ordering and selection operations (e.g., rank correlations and bottom- m selection), and does not rely on semantic structure in the inputs. Moreover, randomly initialized models have independent parameter matrices, suggesting that their outputs under the null exhibit no strong structured dependencies. Therefore, sampling independent Gaussian matrices provides a simple and tractable proxy for modeling the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

Because this construction runs the *same* selection procedure on Gaussian samples, it also reproduces the dependency patterns the pipeline itself introduces (e.g., from selecting $\mathcal{S}$ across dimensions). The agreement is quantitative as well as conceptual: Figure 4 (middle) shows the surrogate distribution matching the empirical null from independently initialized models in centered-Gaussian shape, so the surrogate stands in faithfully for the true null at a fraction of the cost.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

While the Gaussian surrogate captures the behavior of the null distribution under the full pipeline, it still requires repeated sampling and introduces randomness in the estimation process. We therefore also propose the most efficient analytical approximation.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization introduces dependencies across dimensions. Yet Figure 4 shows the analytical Gaussian tracking the surrogate-pipeline distribution closely and consistently—across varying numbers of random inputs n and output dimensions D —so these dependencies are weak enough to leave the approximation intact. The likely mechanism is the relatively high temperature ($T = 10$), which smooths the outputs and suppresses inter-dimension coupling.

²As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

Together, these results justify the use of the analytical Gaussian null in Section 4 as an efficient, deterministic, and accurate approximation, eliminating the need for costly empirical baseline construction.

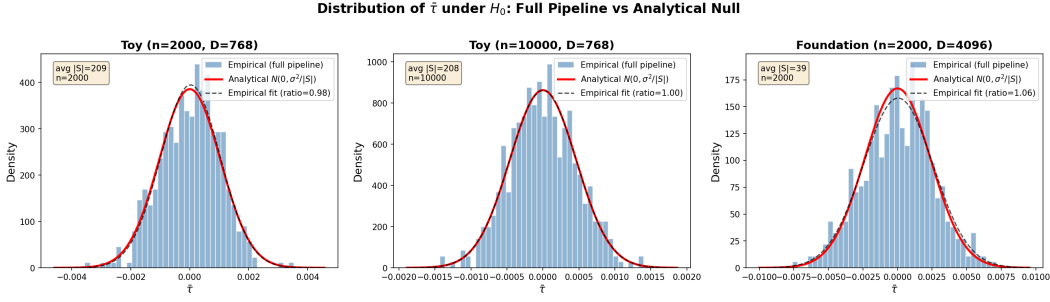


Figure 4: Validation of the analytical Gaussian null. Running the full pipeline (identity-dimension selection and correlation aggregation) on Gaussian surrogate outputs yields a distribution of $\bar{r}$ that lies essentially on top of the analytical Gaussian $\mathcal{N}(0, \sigma^2/|S|)$. The close overlap confirms the analytical approximation captures the null faithfully despite the pipeline’s weak dependencies—justifying the fast, deterministic test used in the main paper.

On variance scaling. The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This mismatch does not affect our hypothesis test results. Our primary evaluation for simulation-based null in Section D.2 relies on the Mann–Whitney U-test, which is rank-based and thus invariant to any positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$. Therefore, all rank statistics and rejection thresholds remain unchanged. For tests that do depend on absolute variance, we have additionally evaluated variance-sensitive statistics such as the t -test, whose statistic scales as $t' = t/c$ when variance decreases. Since $c < 1$ in our case, scaling reduces the magnitude of t and makes the test more conservative rather than inflating significance. Empirically, across extensive experiments in Section D.2, we find that the variance scaling does not affect the power of the t -test, indicating that the distributional separation between the alternative hypothesis and the null is substantially larger than the scaled variance.

For the analytical null used in the main paper, the variance is explicitly included in the normalization of the z -score. Therefore, any scale discrepancy is absorbed in the denominator. Since the analytical variance is slightly larger than the empirical one, the resulting z -scores are smaller, making the test conservative and preventing false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

Our goal is to identify the output dimensions that the initialized model is least likely to predict under random inputs. These dimensions correspond to directions that the model inherently disfavors due to initialization-induced bias (not dependent on the input distribution), forming the basis of a stable fingerprint signal.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings as in the main paper). For each output dimension $j \in [d_{\text{out}}]$, $g_j(x)$ is thus a random variable and its population mean is $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we aim to capture correspond to the m output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap measures how well-separated the target set $\mathcal{M}$ is from the remaining coordinates. Importantly, for a fixed m and input distribution, $\Gamma(m)$ depends solely on the model’s inherent initialization bias. *However,*

increasing the length of random input sequences amplifies this bias via concentration, making the separation more pronounced at the population level.

There are two natural empirical strategies to extract such population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates with the smallest value. Following the notation in Section 4, where g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select the smallest-value coordinate and then take the m most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both approaches are intuitively reasonable. However, as we show below, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, making it inherently noise-sensitive.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first show that the sample-mean-based operator, which selects the m smallest coordinates in the empirical mean vector, is stable under small perturbations to model outputs. The perturbations represents realistic variations in outputs from the *same* model, e.g., numerical noise, or evaluation under different precision modes.

Formally, we model perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ represents stochastic deviations. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma provides a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decreases exponentially with the number of averaged samples n , due to variance reduction from σ^2 per sample to σ^2/n in the empirical mean.

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special case of this assumption.

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\hat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

For any coordinate j , by definition of $\hat{g}'$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \right\}.$$

Equivalently, if $\max_j |\hat{g}'_j - \hat{g}_j| < \hat{\Gamma}(m)/2$, then the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\hat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \hat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \hat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \hat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\hat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\hat{\Gamma}(m)}{2}) \geq \hat{\Gamma}(m) - \hat{\Gamma}(m) = 0.$$

Thus no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$ under this deviation bound, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Since $\hat{\Gamma}(m)$ is fixed under conditioning on X , plugging $\varepsilon = \hat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \hat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\hat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator on each input. Thus, arbitrarily small perturbations to outputs can change decisions with non-vanishing probability. We still consider the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting does not benefit from variance reduction: each input incurs a direct hard decision based on noisy outputs. Increasing n only increases the number of opportunities for index-flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, note that the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of the dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$ (i.e., only a constant number of votes), then a change in $j^*(x_i)$ for a single input x_i is sufficient to swap the ordering across this boundary and thus alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now analyze how many random inputs are required for the empirical mean estimator to recover the true bias. We reuse notation from **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ represents the sorted means in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ is the separation margin. We estimate $\bar{g}$ using n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimensions set is $\mathcal{M}$ with size m and our empirical estimation gives $\hat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\hat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter nor inside index leave the bottom- m set. $\square$

We next interpret this recovery condition in terms of hyperparameters. We again adopt the same sub-Gaussian assumption as in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting, each $g_j(x)$ corresponds to a logit or final hidden activation, which are typically bounded or light-tailed in both initialized networks (due to controlled parameter initialization) and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\hat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\hat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: Cross-seed separability across model families (simulation-based null). Every seed pair yields $p > 0.01$ under both output types and both tests, so distinct seeds are never conflated.

(a) LLaMA-style models					(b) Qwen-style models				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>		<i>t-test</i>	<i>U-test</i>	<i>t-test</i>	<i>U-test</i>
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style) Trained models inherit their initialization’s fingerprint: SeedPrints confirms all four init-to-base pairs at $p < 0.01$ under every output/test combination (down to $4.00\text{e}-10$), while all four baselines miss every pair.

Model Pair	Logits Output		Hidden State		Baselines			
	<i>t-test</i>	<i>u-test</i>	<i>t-test</i>	<i>u-test</i>	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	$3.33\text{e}-3$ ✓	$1.02\text{e}-3$ ✓	$2.20\text{e}-8$ ✓	$6.28\text{e}-8$ ✓	-0.021 ×	0.375 ×	0.580 ×	0.196 ×
s_{123}^{init} vs. s_{123}^{base}	$2.06\text{e}-3$ ✓	$7.33\text{e}-3$ ✓	$7.09\text{e}-6$ ✓	$1.37\text{e}-5$ ✓	0.149 ×	0.369 ×	0.581 ×	0.188 ×
s_{1000}^{init} vs. s_{1000}^{base}	$2.44\text{e}-3$ ✓	$4.14\text{e}-3$ ✓	$5.58\text{e}-4$ ✓	$2.81\text{e}-3$ ✓	-0.252 ×	0.381 ×	0.581 ×	0.188 ×
s_{2000}^{init} vs. s_{2000}^{base}	$5.63\text{e}-3$ ✓	$6.76\text{e}-3$ ✓	$4.00\text{e}-10$ ✓	$1.27\text{e}-9$ ✓	-0.337 ×	0.331 ×	0.581 ×	0.188 ×

Therefore, accurate recovery requires $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$, meaning the number of random queries grows logarithmically with the output dimension, but scales inversely with the squared separation margin γ . The margin γ itself has two contributing factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of m .** Different choices of m correspond to different parts of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$, thus yielding different effective margins in practice.

The former can be controlled by the input length l , while the latter is selected empirically based on the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models with the Hugging Face Transformers Trainer (Wolf et al., 2020), using Accelerate (Gugger et al., 2022) for distributed runs. All open-source models are loaded from their official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License, and other models under Apache-2.0. All datasets are downloaded via the Hugging Face Datasets library (the library is Apache-2.0); dataset content follows each dataset’s stated license.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section provides the complementary results for Section 5, re-running every main-paper comparison under the empirical simulation-based null in Algorithm 1 with both *logits* and *hidden state* outputs and under both the one-sided *t*-test and the Mann–Whitney *U* test. The purpose is robustness: across all four (output $\times$ test) combinations, the accept/reject verdicts match those of the analytical null in the main paper, so none of our conclusions hinges on a particular null construction or test statistic.

Table 11: (Qwen-style) Same conclusion as the LLaMA-style case: all four init-to-base pairs are confirmed at $p < 0.01$ under both tests, with hidden-state p -values reaching $1.06e-24$.

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: Persistence under continual training on divergent corpora (base: seed 1000, openwebtext); U -test is the Mann–Whitney U test. On the far-domain corpus `the_stack`, baselines such as REEF misclassify the true descendant ($0.557/0.313$ for LLaMA/Qwen), whereas SeedPrints stays correct on all cases and reaches p -values down to 10^{-287} .

(a) LLaMA-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0 [✓]	0 [✓]	0 [✓]	7.77e-89 [✓]	1.000 [✓]	0.759 [×]	0.999 [✓]	0.996 [✓]
TinyStories (123)	1.000 [✓]	1.00 [✓]	0.943 [✓]	0.902 [✓]	0.950 [×]	0.658 [✓]	0.332 [✓]	0.012 [✓]
the_stack (1000)	0 [✓]	1.73e-287 [✓]	0 [✓]	3.09e-69 [✓]	0.489 [×]	0.557 [×]	0.585 [×]	0.123 [×]
the_stack (123)	0.616 [✓]	0.479 [✓]	0.732 [✓]	0.831 [✓]	0.445 [✓]	0.580 [✓]	0.301 [✓]	0.026 [✓]

(b) Qwen-style models

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09 [✓]	1.92e-07 [✓]	8.49e-214 [✓]	5.09e-71 [✓]	1.000 [✓]	0.957 [✓]	0.999 [✓]	0.996 [✓]
TinyStories (123)	0.434 [✓]	0.433 [✓]	0.256 [✓]	0.065 [✓]	0.913 [×]	0.199 [✓]	0.328 [✓]	0.039 [✓]
the_stack (1000)	0 [✓]	4.16e-237 [✓]	1.16e-211 [✓]	2.30e-76 [✓]	0.999 [✓]	0.313 [×]	0.995 [✓]	0.976 [✓]
the_stack (123)	0.999 [✓]	0.993 [✓]	0.610 [✓]	0.491 [✓]	0.916 [×]	0.255 [✓]	0.328 [✓]	0.038 [✓]

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

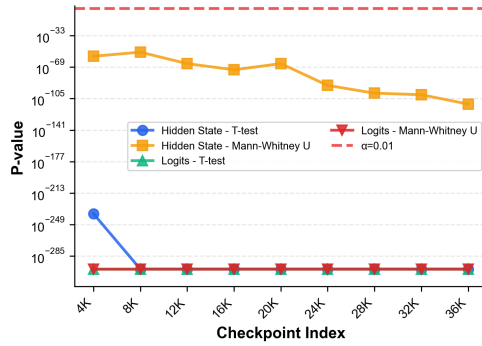
We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Since the simulation-based null is stochastic, we average p -values over 10 independent trials at a significance level of $\alpha = 0.01$. One reading convention matters: below the numerical/sampling floor (e.g., $p < 10^{-20}$), the exact magnitude is not evidentiary—a value like 10^{-260} is not “stronger” than 10^{-20} ; both simply reject the null decisively. We therefore read such entries as clear rejections rather than comparing their magnitudes.

Different initialization seeds produce distinct fingerprints. Table 9a (LLaMA-style) and Table 9b (Qwen-style) report cross-seed p -values (seeds 42, 123, 1000, 2000) under both the t -test and U -test. The result is unanimous: every entry exceeds 0.01 in both tables, so not one distinct-seed pair is falsely merged under either test. Distinct seeds thus yield distinct fingerprints, and models are separable “at birth” independently of the null construction.

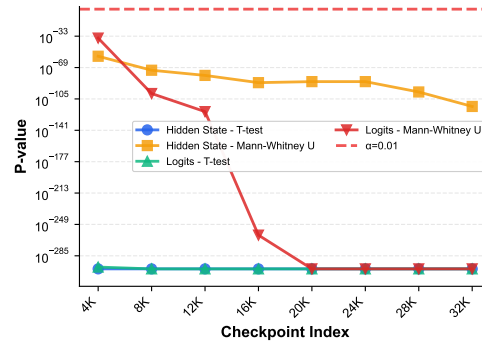
Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) pair each initialization s^{init} with its own descendant s^{base} trained on OpenWebText (Gokaslan et al., 2019) (≈ 10 B tokens). All true-lineage p -values fall well below 0.01—down to $2.20e-8$ (LLaMA, hidden state) and $1.06e-24$ (Qwen, hidden state)—so every descendant is correctly attributed to its initialization; training does not erase the fingerprint. The same table lays the contrast bare: on these identical pairs, all four baselines in Table 10 misfire, confirming once more that their separability tracks training-induced artifacts rather than initialization lineage.

Table 13: Seed remains the sole determinant even under the simulation-based null: with corpus and data order fixed, every cross-seed pair stays at $p > 0.01$ across both output types and both tests, for LLaMA- and Qwen-style models alike.

(a) LLaMA-style models					(b) Qwen-style models				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style models



(b) Qwen-style models

Figure 5: All-stage verification: along the full controlled pretraining trajectory, SeedPrints keeps $p < 0.01$ at every checkpoint for both LLaMA-style and Qwen-style models—no early-stage misses.

Identical data and order do not make fingerprints converge. In Table 13a (LLaMA-style) and Table 13b (Qwen-style), all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ share *exactly the same corpus (OpenWebText) and data order* (the training seed is fixed to lock the curriculum), differing only in initialization seed i —the strongest test of whether identical data can wash out the fingerprint. It does not: every cross-seed p -value stays above 0.01 under both output types and both tests, standing in sharp contrast to the near-zero true-lineage values of Table 10. The fingerprint remains seed-specific even under identical data and curriculum.

Continual training on diverse datasets does not confound the fingerprint. The seed-level experiments above establish SeedPrints as a biometric fingerprint; from a copyright standpoint, a baseline’s inability to separate seeds is not itself a flaw, since seeds carry no clear copyright status. The consequential and measurable failure is that baseline attribution is *misled by data distribution, losing the true lineage when the training distribution shifts substantially during continued training*.

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two deliberately dissimilar corpora—TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code)—and ask each method to pick the true descendant over a *distractor* derived from a different base (seed 123, different data order on OpenWebText, then continued on the *same* corpus).

The split is once again by domain. On the far-domain corpus The Stack, the baselines fail to recover the true descendant (e.g., REEF 0.557/0.313 for LLaMA/Qwen, both below 0.8), a clear signature of tracking *domain similarity* rather than lineage: TinyStories stays near the OpenWebText distribution while The Stack diverges enough to evade detection. SeedPrints instead attributes lineage correctly on both corpora and under both tests—returning p -values as small as 10^{-287} (logits) and 10^{-211} (hidden state) for the true descendants—so its fingerprint is no proxy for data distribution: it survives a large domain shift and persists well beyond initial pretraining.

Table 14: Cross-size comparison between Qwen-2.5-14B (target) and smaller same-family suspects. SeedPrint p -values all sit above $\alpha=0.01$ (0.39–0.97), correctly declining to flag any smaller model as a descendant, whereas REEF scores cluster in a narrow 0.825–0.883 band and HUREF scores hover near zero, neither offering a clean cutoff.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We run a controlled OpenWebText experiment with known initialization seed, treating each intermediate checkpoint as the base and testing whether our method recovers its offspring along the same trajectory. Every variant—both output types and both tests—keeps its p -value below the 0.01 threshold at all checkpoints (Figure 5a for LLaMA-style, Figure 5b for Qwen-style), so detection is stable end-to-end under continuous optimization, with no weak point along the curve.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To test whether SeedPrints “collapses” to a single family-level signature as parameter size varies, we measure cross-size agreement between Qwen-2.5-14B (target) and smaller members of the Qwen-2.5 family. Table 14 reports SeedPrint ($\uparrow$) alongside two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$), so the three methods’ behavior can be compared directly on the same suspects.

In this cross-size study within the Qwen-2.5 series, Qwen-2.5-14B is the protected target and Qwen-2.5-{0.5B, 1.5B, 3B, 7B} are suspects: same architectural family, but *not* descendants of the 14B model, so the correct verdict for every suspect is “distinct.” As in the main paper, SeedPrints returns a p -value for the hypothesis “the suspect shares the target’s initialization-born fingerprint,” whereas REEF and HUREF emit continuous similarity scores that leave the accept/reject cutoff to the user.

Three points stand out. First, every SeedPrints p -value clears the significance level ($\alpha=0.01$)—the four values span 0.3898 to 0.9707—so no smaller Qwen-2.5 model is flagged as sharing lineage with the 14B target. This is exactly the desired outcome: despite a shared family and architecture, the test does not collapse to one family-level signature that would spuriously conflate distinct pre-training runs.

Second, those four p -values are spread across the $[0, 1]$ range rather than piling up at a single value. We read these magnitudes not as a continuous similarity score but as inputs to a calibrated yes/no decision; still, the spread shows the statistic remains sensitive to the specific architecture-plus-seed combination instead of degenerating into a generic Qwen-2.5 pattern.

Third, the similarity baselines offer no comparable clarity. REEF stays in a narrow 0.8250–0.8833 band across all four sizes—straddling the usual 0.8 cutoff—while HUREF returns values on the order of 10^{-5} . With no built-in decision rule, an operator must hand-pick a threshold, and no single cutoff cleanly separates same-family cross-size variants from genuine descendants. SeedPrints, by contrast, delivers a direct statistical decision at a prescribed significance level, giving an operationally clearer answer to “do these two models plausibly share a training lineage?”

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family. For each derived model we also indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” refer to *pre-trained* and *instruction-tuned* source models, respectively.

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Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2B-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation relative to the PT/IT source model: *finetune* includes supervised/preference optimization variants; *adapter* includes LoRA/QLoRA-style modules; *merge* includes model soups and TIES-style merges; *quantization* includes GPTQ/INTx variants; *distillation* includes student models distilled from reasoning-augmented teachers.

1566 E.2 SAMPLE RESULTS 1567

1568 Table 15 shows partial results from one LeafBench run (u -test, simulation-based null), reporting
1569 $1 - p$ as the similarity score at significance level 0.01. The separation is sharp enough that LeafBench
1570 recovers an optimal decision threshold of 0.9920—i.e., same-lineage pairs concentrate above it and
1571 unrelated pairs below—further evidence that the test yields a clean, thresholdable decision boundary
1572 in practice.

1573 The key pattern to read from the matrix is distributional, not per-cell: although an individual p -
1574 value is not a linear similarity, same-lineage pairs collapse to markedly low p -values (high $1 - p$)
1575 while unrelated pairs stay near uniform, producing the well-separated two-population structure the
1576 highlighted cells make visible.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample results.** Pairwise scores ($1 - p$) between target models (rows) and base models (columns); **green** marks same-lineage cells with $1 - p \geq 0.99$. The green mass concentrates on the diagonal blocks (each family with its own base), while off-block entries stay low—the two-population separation that underlies the strong per-family AUC/KS scores.